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Surname:															
First Name(s):															

Department of Mathematics and Applied Mathematics
MAM2000W: 2LA (Linear Algebra)

Test 1

14 March 2022

Time : 1.5 hours

Maximum marks: 50

Full marks: 50

- This question paper consists of 9 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Basic scientific calculators are allowed.
- For Section A, only final answers will be marked. For Section B, full answers must be given.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	TOTAL
3	6	8	9	6	13	5	50

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Section A

In this section, give your final answer only. Do not show your working.

Question 1. [3 points] Let A be a 3×3 skew-symmetric matrix such that $a_{13} = -2$, $a_{21} = 5$ and $a_{23} = 7$. Find the matrix $B = A - 3I_3$.

$$B = \begin{pmatrix} -3 & -5 & -2 \\ 5 & -3 & 7 \\ 2 & -7 & -3 \end{pmatrix}$$

Question 2. [6 points] Let $\mathcal{U}$ denote the set of all upper triangular, $\mathcal{L}$ the set of all lower triangular, $\mathcal{I}$ the set of all invertible, $\mathcal{S}$ the set of all symmetric and $\mathcal{E}$ the set of all elementary 3×3 matrices. From the list of matrices

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 2 & 0 \\ 5 & 6 & 3 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix},$$

$$D = \begin{pmatrix} 1 & 0 & -1 \\ 4 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad E = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \quad F = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

choose an element of each of the sets in (a), (b), (c), and (d) below, if possible. Write only one of A , B , C , D , E , F in the appropriate block below. If it is impossible to find the corresponding matrix, write "Impossible". You may use a matrix more than once, or not at all.

(a) $\mathcal{L} \cap \mathcal{U}$

1 mark

F

(b) $\mathcal{L} \cap \mathcal{E}$

1 mark

C

(c) $\mathcal{S} \cap \mathcal{I}$

2 marks

E

(A is not invertible)

(d) $\{A : A^T \in \mathcal{U} \cap \mathcal{I} \text{ and } A \notin \mathcal{E}\}$

2 marks

B

(F is not invertible, C is elementary)

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Question 3. [8 points] The augmented matrix of a system of linear equations is

$$\left(\begin{array}{cccc|c} 3 & 0 & 0 & 0 & -6 \\ 0 & 2 & 0 & 0 & 7 \\ 0 & 0 & k & k(k-1) & k^2-1 \\ 0 & 0 & 0 & k+1 & 0 \end{array} \right),$$

where k is a constant. Give the solution set for this system in the case where $k = -1$.

3 marks

$$\left\{ \begin{pmatrix} -2 \\ 3.5 \\ 2.5 \\ s \end{pmatrix} : s \in \mathbb{R} \right\}$$

Now give **all** the values of k , if any, for which the system has

(a) no solution;

2 marks

$$K = 0$$

(b) infinitely many solutions;

2 marks

$$K = -1$$

(c) a unique solution.

1 mark

$$K \neq 0 \text{ and } K \neq -1$$

If there is no value of k in any of the cases above, write "No k ".

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Section B

In this section, give full answers to all the questions. Show all your working.

Question 4. [9 points] Let $A = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 3 & 0 \\ -2 & 1 & 0 \end{pmatrix}$.

(a) Find elementary matrices E_1 , E_2 and E_3 and a matrix U in row echelon form such $E_3 E_2 E_1 A = U$.

4 marks

$$\begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & -1 \\ -2 & 1 & 0 \end{pmatrix} \quad R_1 \leftrightarrow R_2 \quad E_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & -1 \\ 0 & 7 & 0 \end{pmatrix} \quad R_3 \rightarrow R_3 + 2R_1 \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 7 \end{pmatrix} \quad R_3 \rightarrow R_3 - 7R_2$$

U

$$E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -7 & 1 \end{pmatrix}$$

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(b) Is A invertible? Use only your answer to (a) to give a reason.

2 marks

A is invertible because every column in U has a pivot (each entry on the diagonal in U is non zero).

(c) Use your answer to (a) to find two 3×3 matrices B and C such that neither is equal to the identity matrix and $A = BC$. Show how you arrived at your answer.

3 marks

$$B = E_1^{-1} \cdot E_2^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

There are of course many different correct answers, corresponding to how you choose the "factors" of A using the expression in (a).

$$C = E_3^{-1} U = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & -1 \\ 0 & 7 & 0 \end{pmatrix}$$

Question 5. [6 points] Prove or disprove:

(a) The product of two elementary matrices is an elementary matrix.

2 marks

This is false.

A counterexample is for instance

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ -3 & 1 \end{pmatrix}$$

↑
elementary

↑
elementary

↑
not elementary

Of course, there are many other counter-examples.

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- (b) Every upper triangular matrix is in row echelon form.

2 marks

This is false. A counterexample is for instance the matrix

$$\begin{pmatrix} 0 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 6 \end{pmatrix}.$$

Of course, there are many other counter-examples.

- (c) A system of m linear equations in n variables has infinitely many solutions if $m < n$.

2 marks

This is false.

A counterexample is for instance the system

$$x_1 + x_2 + x_3 = 1$$

$$x_1 + x_2 + x_3 = 2.$$

This system is inconsistent.

Of course, there are many other counter-examples.

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Question 6. [13 points] Consider the following statement:

For all $n \times n$ matrices A and B , if AB is invertible, then A and B are invertible. (*)

(a) Write down the converse of the statement (*).

1 mark

For all $n \times n$ matrices A and B , if A and B are invertible then AB is invertible.

(b) Write down the contra-positive of the statement (*)

1 mark

For all $n \times n$ matrices A and B , if A is not invertible or B is not invertible then AB is not invertible.

Read the following proof of the statement (*) and then answer the questions following it.

Suppose that B is not invertible. (1)

Then there exists a non-zero vector $\mathbf{x}$ such that $B\mathbf{x} = \mathbf{0}$ (2)

But then $(AB)\mathbf{x} = A(B\mathbf{x}) = A\mathbf{0} = \mathbf{0}$, (3)

which shows that AB not invertible. (4)

Thus if AB is invertible, then so is B .

But then B^{-1} is invertible. (5)

However if both AB and B^{-1} are invertible, then so is $(AB)B^{-1}$ (6)

Hence A is invertible.

(c) Which statement is proved in lines (1) – (4)?

1 mark

If B is not invertible then AB is not invertible.

(d) Explain carefully how (2) follows from (1). What property of the matrix B are we using?

1 marks

If B is not invertible then B is singular.
Then by definition, there exists a vector $\mathbf{x} \neq \mathbf{0}$
s.t. $B\mathbf{x} = \mathbf{0}$.

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(e) How does it follow from line (3) that AB is not invertible?

2 marks

In (3), it is shown that there exists $\underline{x} \neq 0$ s.t.
 $(AB)\underline{x} = 0$. Hence by definition, AB is singular
 and therefore not invertible.

(f) Use the definition of invertibility to show why we can say in line (5) that B^{-1} is invertible.

2 marks

Since by definition, B^{-1} is the inverse of B ,
 it holds that

$$B^{-1} \cdot B = B \cdot B^{-1} = I.$$

By definition, it follows that B is the inverse
 of B^{-1} . Hence B^{-1} has an inverse and is therefore
 invertible.

(g) Give a proof of the converse of the statement (*).

3 marks

If A is invertible, A has the inverse A^{-1} .
 If B is invertible, B has the inverse B^{-1} .
 Then we have

$$AB \cdot (B^{-1}A^{-1}) = (B^{-1}A^{-1})(AB) = I.$$

$$I = A \overset{||}{(BB^{-1})} A^{-1}$$

$$B^{-1} \overset{||}{(A^{-1}A)} B = I$$

Hence $B^{-1}A^{-1}$ is
 the inverse of
 AB . Therefore, AB
 is invertible.

(h) Is this converse used anywhere in the proof above? If so, where is it used, and how?

2 marks

The converse is used in line 6
 where it is shown that $(AB)B^{-1}$ is invertible
 because AB and B^{-1} are invertible.

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Question 7. [5 points] A square matrix is called a *permutation matrix* if each row and each column contains exactly one entry equal to 1, and all the other entries are 0. Prove or disprove each of the following statements:

- (a) Each permutation matrix P is invertible, and $P^{-1} = P^T$.

3 marks

True. Let P be an $n \times n$ permutation matrix. Since each row of P only contains one non-zero entry, namely a 1, a finite number of row interchanges will transform P into I_n . Hence P is row-equivalent to I_n and therefore invertible (by the Invertibility Theorem).

Let the elementary matrices corresponding to the row interchanges be $E_1, E_2, \dots, E_k$, and $E_k \cdots E_2 E_1 P = I_n$. Then $P = E_1^{-1} \cdots E_k^{-1} = E_1 \cdots E_k$, since $E_i^{-1} = E_i$ for $i = 1, 2, \dots, n$. If $E = (e_{ij})$ is the elementary matrix corresponding to the row operation of interchanging row k and row m , then $e_{mk} = e_{km} = 1$, $e_{ii} = 1$ if $i \neq m, k$ and $e_{ij} = 0$ otherwise. It follows that E is symmetrical, so $E^T = E$. From this it follows that

$$P^{-1} = E_k \cdots E_1, \quad P^T = (E_1 \cdots E_k)^T = E_k^T \cdots E_1^T = E_k \cdots E_1 = P^{-1}.$$

- (b) There is more than one upper triangular $n \times n$ permutation matrix.

2 marks

False. For fixed n , I_n is an upper triangular permutation matrix. If there is another $n \times n$ upper triangular permutation matrix $P = (p_{ij})$, then for some $i \neq j$, $p_{ij} = 1$. If $i > j$, this will contradict the fact that for an upper triangular matrix $p_{ij} = 0$ for $i > j$. If $i < j$, column j of P will contain two 1's, contradicting the fact that P is a permutation matrix. Hence I_n is the only $n \times n$ permutation matrix.

Notes

- (a) In this question the way you presented your solution was very important; you did not get marks for a confused, difficult to understand answer, or making claims without justification.
- (b) The solutions given above are not the only correct ones. In (a), for example, you could also have checked directly, using the definition of a permutation matrix, that for any permutation matrix P , $PP^T = I = P^T P$, and concluded that $P^{-1} = P^T$.
- (c) Checking an example is not a proof of a general statement. If you gave an example of a permutation matrix P in (a) and checked that for this example $P^{-1} = P^T$, you did not get any marks.
- (d) Row equivalence is not the same as equality. You cannot argue that since P is row equivalent to I_n and $I_n^{-1} = I_n^T$ that also $P^{-1} = P^T$.
- (e) A permutation matrix need not be equal to the identity matrix or an elementary matrix corresponding to a row interchange. For $n > 3$, an $n \times n$ matrix could be obtained from I_n by making more than one row interchange.